

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE CONTINUOUS ASSESSMENT TEST - II - MARCH-2026 CET 05101 OPERATING SYSTEM

SEMESTER: I

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

CURATED ANSWERS PROVIDED BY 4LAZIE - SMART IN BRAIN

PART A: SHORT ANSWER QUESTIONS

Q1. What are the drawbacks of Contiguous Allocation of Disk Space?

Answer:

- **External Fragmentation:** As files are allocated and deleted, free disk space is broken into small pieces.
- **Difficulty in finding space:** If a file needs to grow later, there might not be enough adjacent space available, leading to allocation failure.

Q2. Give a real-life example of deadlock.

Answer: A classic real-life example is two trains approaching each other on a single track. Neither can move forward until the other moves backward, resulting in a permanent standstill (deadlock).

Q3. What is process synchronization?

Answer: Process synchronization is the coordination of execution of multiple processes in a multi-tasking system to ensure that they access shared resources in a mutually exclusive manner, preventing data inconsistency.

Q4. Define semaphores and gives two types of semaphores

Answer: A semaphore is an integer variable used to control access to a common resource by multiple processes in a concurrent system to avoid critical section problems.

- **Counting Semaphore:** Can range over an unrestricted integer domain.
- **Binary Semaphore:** Can only be 0 or 1, also known as mutex locks.

Q5. What is deadlock ignorance?

Answer: Deadlock ignorance is a strategy (used by operating systems like UNIX and Windows) where the OS simply ignores deadlocks, assuming they will rarely happen. If a deadlock occurs, it is left up to the user to manually handle it (e.g., by restarting the system). This is also known as the Ostrich Algorithm.

Q6. Define critical section problem.

Answer: The critical section problem is designing a protocol that processes can use to cooperate. A critical section is a segment of code where a process accesses shared variables. The problem ensures that when one process is in its critical section, no other process is allowed to execute in its critical section.

Q7. State two types of fragmentation in memory management

Answer:

- Internal Fragmentation
- External Fragmentation

PART B: EXPLANATORY QUESTIONS

Q8. Elaborate three condition used to solve critical section problem in Operating system.

Answer: To solve the critical section problem, three conditions must be satisfied:

- **Mutual Exclusion:** If a process is executing in its critical section, no other process can execute in theirs.
- **Progress:** If no process is executing in its critical section, the selection of the next process to enter cannot be postponed indefinitely.
- **Bounded Waiting:** A bound must exist on the number of times other processes are allowed to enter their critical sections after a process has made a request to enter its critical section.

Q9. Discuss deeply about four condition that occurs in deadlock in operating system

Answer: Deadlock occurs if these four conditions (Coffman conditions) hold simultaneously:

- **Mutual Exclusion:** At least one resource must be held in a non-sharable mode.
- **Hold and Wait:** A process holding at least one resource is waiting to acquire additional resources held by other processes.
- **No Preemption:** A resource cannot be forcibly taken from a process; it can only be released voluntarily.
- **Circular Wait:** A set of waiting processes exist such that P₀ is waiting for P₁, P₁ is waiting for P₂, and P_n is waiting for P₀.

Q10. Describe fragmentation process in operating system and explain shortly about two types of fragmentation in operating system.

Answer: Fragmentation is a problem in operating systems where memory blocks cannot be allocated to processes because the free memory is broken into small pieces. Types:

- **Internal Fragmentation:** Occurs when memory allocated to a process is slightly larger than the requested memory. The leftover space inside the allocated block is wasted.
- **External Fragmentation:** Occurs when total free memory is enough to satisfy a request, but it is scattered into non-contiguous small blocks.

Q11. Explain in details about thread state in operating system with the aids of diagram.

Answer: A thread can be in any of the following states:

Smart Tip: Note: In the exam, remember to draw a diagram with boxes for the states and arrows showing transitions (e.g., New → Runnable ↔ Running → Terminated).

- **New:** The thread has been created but not yet started.
- **Runnable:** The thread is ready to execute and waiting for CPU time.
- **Running:** The thread is currently executing on the CPU.
- **Blocked/Waiting:** The thread is waiting for an event (like I/O) to complete.

- **Terminated/Dead:** The thread has finished execution.

Q12. Briefly discuss four strategies used to handle deadlock in operating system.

Answer:

- **Deadlock Ignorance:** Ignore the problem altogether (Ostrich Algorithm) and let the user handle it.
- **Deadlock Prevention:** Restrict resource requests to ensure at least one of the four Coffman conditions cannot occur.
- **Deadlock Avoidance:** The OS requires additional information to dynamically evaluate resource requests and ensure the system remains in a safe state.
- **Deadlock Detection and Recovery:** Allow deadlock to occur, detect it, and recover by killing processes or preempting resources.

Q13. Summarize shortly about two types of memory management in operating system..

Answer:

- **Contiguous Memory Management:** Each process is contained in a single contiguous block of memory. It is simple but suffers from external fragmentation.
- **Non-Contiguous Memory Management (Paging/Segmentation):** A process is divided into smaller parts and stored in different, non-adjacent locations in physical memory, which solves external fragmentation.

PART C: EXPLANATORY QUESTIONS

Q14. Suppose the user process's size is 4096KB and is a standard hard disk where swapping has a data transfer rate of 1Mbps. Determine how long will take to transfer from main memory to secondary memory then calculate total transfer rate taken in swap in and swap out the data processed in a memory

Answer: Given:

- Process Size = 4096 KB = 4 MB
- Transfer Rate = 1 MBps (Assuming Megabytes per second as standard in OS textbooks)

Calculations:

1. Transfer time (Swap Out to secondary memory) = Size / Transfer Rate
= 4 MB / 1 MBps = 4 Seconds.
2. Total transfer time (Swap In and Swap Out) = Time to Swap Out + Time to Swap In
= 4 Seconds (Out) + 4 Seconds (In) = 8 Seconds.

Smart Tip: If the transfer rate was literally 1 Mbps (Megabits), you would convert 4MB to Megabits ($4 \times 8 = 32$ Megabits). The time would then be 32 seconds out + 32 seconds in = 64 seconds total.

Q15. Explain in details about three multithreading model in operating system with the help of diagrams.

Answer: Multithreading models define the relationship between user-level threads and kernel-level threads:

Smart Tip: In the exam, draw circles for user threads mapping to squares representing kernel threads to illustrate the models.

- **Many-to-One Model:** Maps many user-level threads to a single kernel thread. It's efficient, but if one user thread

blocks on I/O, the entire process blocks.

- **One-to-One Model:** Maps each user thread to its own kernel thread. It allows true concurrency, but creating too many kernel threads can burden system performance.
- **Many-to-Many Model:** Multiplexes many user-level threads to a smaller or equal number of kernel threads. It balances concurrency and system efficiency perfectly.

Q16. Briefly explain in details about three major activities of an operating system with regard to memory management?

Answer: The OS acts as the manager of the system's memory. Its major activities include:

- **Tracking memory usage:** Keeping track of which parts of memory are currently being used and by which processes.
- **Memory transfer:** Deciding which processes (or parts of processes) and data to move into and out of memory when space becomes available.
- **Allocation and Deallocation:** Allocating memory space as needed to new processes and deallocating memory space when a process terminates.

Q17. Elaborate three steps required to handle a page fault in demand paging in operating system.

Answer: A page fault occurs when a program tries to access a page that is mapped in address space but not loaded in physical memory. The three steps to handle it are:

- **Trap to the OS:** The OS checks the internal page table to determine if the reference was a valid memory access. If valid, but the page is missing, a page fault is triggered.
- **Bring Page into Memory:** The OS finds a free frame in physical memory and schedules a disk operation to read the desired page from the disk into the newly allocated frame.
- **Restart Instruction:** Once the disk read is complete, the OS updates the page table to indicate the page is now in memory, and restarts the instruction that caused the page fault.